

Quantum Gravitation

Lecture 12

Quantum States of Dense Media

- Degrees of freedom
- Quantization
- Spectrum and response
- Consistency checks
- Example

■ Degrees of freedom

We separate propagating polarizations from constrained components before quantization.

$$\omega_0^2(M_c) = 0, M_c = 1.9M_\odot$$

■ Quantization

Canonical normalization is fixed before creation and annihilation operators are introduced.

$$\Psi = \langle a_{k \approx 0, s} \rangle / \sqrt{V}$$

■ Spectrum and response

Poles, residues, and group velocity are read from the same quadratic operator.

$$V_{eff} = a|\Psi|^2 + b|\Psi|^4 + \dots$$

■ Consistency checks

Hermiticity, positive norm, source conservation, and the low-energy limit are checked explicitly.

■ Example

A short numerical estimate fixes the scale of the effect before a full fit is attempted.

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Reading: Dense Medium Quantum States, ch. 2; D-block registry; Seo, ch. 12